

Exercise 1

Consider the sets of natural numbers $A = \{1, 2, 3\}$ and $B = \{2, 3\}$. Which of the following statements is true? (Justify your answers!)

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|---|---|
| 1. $A \subseteq B$; | 7. $\{3\} \in A \cap B$; |
| 2. $B \subseteq A$; | 8. $(1, 2) \in A \times B$; |
| 3. $A \subset \mathbb{N}$; | 9. $(2, 1) \in A \times B$; |
| 4. $a \in A$; | 10. $(1, 2, 3) \in A^3$; |
| 5. $\emptyset \in \mathcal{P}(A)$; | 11. $A \cap B \in \mathcal{P}(A)$; |
| 6. $\{\emptyset\} \in \mathcal{P}(A)$; | 12. $\overline{A} \in \mathcal{P}(A)$. |

Solution of Exercise 1

1. False. Because $1 \notin B$;
2. True. Because $2, 3 \in A$;
3. True. Because $1, 2, 3 \in \mathbb{N}$ (are natural numbers) and $A \neq \mathbb{N}$;
4. False. Because the elements of A are only 1, 2, and 3;
5. True. Because the empty set $\emptyset$ is a subset of any set, and so, it is of A ;
6. False. Because $\{\emptyset\}$ is the singleton set containing $\emptyset$, which is not a subset of A .
7. False. Because $\{3\}$ is the singleton set containing 3, which is neither a member of A nor a member of B ;
8. True. Because $1 \in A$ and $2 \in B$.
9. False. Because $1 \notin B$;
10. True. Because $1, 2, 3 \in A$ and $A^3 = A \times A \times A$;
11. True. Because, as all the elements in $A \cap B$ must also be elements of A , then $A \cap B \subseteq A$;
12. False.

Justification 1: By definition of complement of a set, $\overline{A}$ does not contain elements in A . Thus, $\overline{A}$ is not a subset of A .

Justification 2: $\overline{A} = \{0, 4, 5, 6, \dots\}$ is the complement of A as a subset of $\mathbb{N}$ (which, from the description of the exercise, seems to be the domain of discourse). Thus, $\overline{A}$ is not a subset of A , as $0 \notin A$.

Exercise 2

1. Let $A = \{x, y, z\}$ and $B = \{x, y\}$. Provide the element listing description of the following sets:

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|--------------------------------|---------------------------------------|
| (a) $A \cup B$ (union) | (c) $\mathcal{P}(B)$ (powerset); |
| (b) $A \cap B$ (intersection); | (d) $A \times B$ (Cartesian product). |

2. Let A, B, C be sets. Provide a formal (“rule of membership”) description of the following sets (for example, a formal description for the Cartesian product is $A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$):

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|------------------|--|-------------------------------------|
| (a) $A \cup B$; | (c) $\mathcal{P}(A)$; | (e) $(A \cap C) \times B$; |
| (b) $A \cap B$; | (d) $(A \times B) \cap (C \times B)$; | (f) $\mathcal{P}(\mathcal{P}(A))$. |

Solution of Exercise 2

1. The element description of the above sets is given below

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|--------------------------------|---|
| (a) $A \cup B = \{x, y, z\}$; | (c) $\mathcal{P}(B) = \{\emptyset, \{x\}, \{y\}, B\}$; |
| (b) $A \cap B = \{x, y\}$; | (d) $A \times B = \{(x, x), (x, y), (y, x), (y, y), (z, x), (z, y)\}$. |

2. Note that also alternative formulations might be correct.

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| (a) $A \cup B = \{x \mid x \in A \vee x \in B\}$; |
| (b) $A \cap B = \{x \mid x \in A \wedge x \in B\}$; |
| (c) $\mathcal{P}(A) = \{A' \mid A' \subseteq A\}$; |
| (d) $(A \times B) \cap (C \times B) = \{(x, y) \mid x \in A \wedge x \in C \wedge y \in B\}$; |
| (e) $(A \cap C) \times B = \{(x, y) \mid x \in A \wedge x \in C \wedge y \in B\}$; |
| (f) $\mathcal{P}(\mathcal{P}(A)) = \{F \mid F \subseteq \mathcal{P}(A)\}$ |

Exercise 3

Write the following informal statements by using the formal language of first order logic.

1. There exists an integer m such that $m + m$ is greater than 8;
2. Every natural number can be either even or odd;
3. For all $\varepsilon > 0$, there exists a natural number N such that, for all $n \geq N$, $|x - x_n| \leq \varepsilon$.

Solution of Exercise 3

Note that also alternative slightly different formulations might be correct.

1. $\exists m \in \mathbb{Z}. m + m > 8$;
2. $\forall n \in \mathbb{N}. (\exists m \in \mathbb{N}. (n = 2m) \vee (n = 2m + 1))$;
3. $\forall \varepsilon > 0. (\exists N \in \mathbb{N}. (\forall n \geq N. |x - x_n| \leq \varepsilon))$.

Exercise 4

Provide examples of relations that satisfy the following conditions:

1. Reflexive and symmetric but not transitive;
2. Reflexive and transitive but not symmetric;
3. Symmetric and transitive but not reflexive.

Solution of Exercise 4

Let $A = \{x, y, z\}$. The following are binary relations on X :

1. $R_1 = \{(x, x), (y, y), (z, z), (x, y), (y, x), (x, z), (z, x)\}$. It is clearly reflexive and symmetric. It is not transitive, because $(y, x), (x, z) \in R_1$ but $(y, z) \notin R_1$;
2. $R_2 = \{(x, x), (y, y), (z, z), (x, y), (y, z), (x, z)\}$. It is clearly reflexive and transitive. It is not symmetric, because $(x, y) \in R_2$ but $(y, x) \notin R_2$;
3. $R_3 = \{(x, x), (y, y), (x, y), (y, x)\}$. It is clearly symmetric and transitive. It is not reflexive, because $(z, z) \notin R_3$.

Exercise 5

Let $C = \{1, 2, 17, 484\}$ and $D = \{x, y, z\}$ and consider the set of pairs

$$R = \{(1, x), (2, y), (17, y), (17, z)\}.$$

1. Does R encode a function from C to D ?
2. Is R a binary relation between C and D ?

3. Is $R \cup \{484\}$ a relation between C and D ?
4. Let $X \setminus Y$ denote the elements of X not present in Y . With this, does $R \setminus \{(17, z)\}$ encode a function from C to D ? If so, is the function total or partial?

Solution of Exercise 5

1. No, because 17 is mapped to both y and z ;
2. Yes, because $R \subseteq C \times D$;
3. No, because $484 \in R \cup \{484\}$ but $484 \notin C \times D$;
4. Yes, it encodes a function. It is partial because 484 is not mapped to any value in D .

Exercise 6*

Prove that, for any two integers $m, n \in \mathbb{Z}$, if $m + n \geq 19$, then $m \geq 10$ or $n \geq 10$.
(Hint: is it easier to provide a constructive argument or use a proof by contradiction?)

Solution of Exercise 6

We prove the statement using a proof by contradiction. Assume that $m + n \geq 19$, and for the sake of finding a contradiction, that both $m, n < 10$. By the latter assumption, $m + n < 20$. Therefore, it must be that $m + n = 19$. By subtracting 10 from both sides of the equation we get $m + n - 10 = 9$; and since $n - 10$ is a negative quantity (because $n < 10$), we also obtain that $m > 9$. This is a contradiction, as there is no integer number that is both $m > 9$ and $m < 10$.

Exercise 7

Find the error in the following proof that all horses are the same color.

CLAIM: In any set of h horses, all horses are the same color.

PROOF: By induction on h .

Base Case: For $h = 1$. In any set containing just one horse, all horses clearly are the same color.

Inductive Step: For $k \geq 1$, assume that the claim is true for $h = k$ and prove that it is true for $h = k + 1$. Take any set H of $k + 1$ horses. We show that all the horses in this set are the same color. Remove one horse from this set to obtain the set H_1 with just k horses. By the induction hypothesis, all the horses in H_1 are the same color. Now replace the removed horse and remove a different one to obtain the set H_2 . By the same argument, all the horses in H_2 are the same color. Therefore, all the horses in H must be the same color, and the proof is complete.

Solution of Exercise 7

The problem is in the inductive step. Consider the case $h = 2$ and set of horses given by $H = \{\text{black}, \text{white}\}$. Although it is true that $H_1 = \{\text{black}\}$ and $H_2 = \{\text{white}\}$ contain only horses of the same color, the conclusion of H having all horses of the same color fails.

Implicitly, the argument used in the proof assumes that there exists at least another horse that can be used to make a comparison of color between the element missing in H_1 and H_2 . However, this assumption fails in the example above.

Exercise 8*

Prove, by induction on $n \geq 1$ that, $4^n + 14$ is divisible by 6.

Solution of Exercise 8

Let $P(n)$ be the property “there exists integer m such that $4^n + 14 = 6m$ ” (this is *the* definition for a number to be divisible by 6).

Base Case: We prove $P(1)$. This follows because $4^1 + 14 = 6 \cdot 3$.

Inductive Step: For $i \geq 1$ assume $P(i)$ is true. Then, as $4^i + 14 = 6m$, by a simple rearrangement we have that $4^i = 6m - 14$. Next we show that $P(i + 1)$ also follows.

$$\begin{aligned} 4^{i+1} + 14 &= 4(4^i) + 14 \\ &= 4(6m - 14) + 14 && \text{(inductive hp.)} \\ &= 24m - 42 \\ &= 6(4m - 7) . \end{aligned}$$

Indeed, from the above, $m' = 4m - 7$ is the integer showing that $P(i + 1)$ holds.

Exercise 9 (optional)

Prove that there are infinitely many prime numbers.

Solution of Exercise 9

Assume, by way of contradiction, that there are a finite number of primes $p_1, p_2, p_3, \dots, p_n$. Let k be the product of all these primes plus one:

$$k = p_1 p_2 p_3 \cdots p_n + 1.$$

The integer k is clearly greater than 1 and because k is an integer, there must exist some prime number P which divides it. However, P cannot be any one of $p_1, p_2, p_3, \dots, p_n$ because if it were then P could divide the difference $k - p_1 p_2 p_3 \cdots p_n$ and $k - p_1 p_2 p_3 \cdots p_n = 1$.

P cannot divide 1 and also be prime (which is by definition greater than one). This is a contradiction. Therefore our assumption is false so there must be an infinite number of prime numbers.